

Affordances for Harm

*How Offenders Misuse Platform Capabilities
to Exploit Children, and Where to Intervene*

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Corpus	6,323 cases	70,000+ features	56 sources
Coverage	61 task forces	2,864 law enforcement agencies	
Timespan	2002–2026		
Platforms	30+ analyzed		

Abstract

For years, the history of the internet has been championed through the lens of platforms, growth, and innovation. Far less often do we systematically examine the harms that have scaled alongside that same infrastructure.

From large-scale content distribution and global connectivity to encrypted messaging and AI-generated imagery, the capabilities that make technology worth using have also made it exploitable. The same infrastructure that connected billions of people quietly rewired the economics of child exploitation—not because these technologies were designed to cause harm, but because offenders are adaptive: some pursue children deliberately, others are capitalizing on access and opportunity the internet has made possible for the first time.

This paper draws on 6,323 curated Internet Crimes Against Children case records spanning 2002 to 2026 and asks: which platform capabilities offenders exploit most consistently, how technology is weaponized within specific offense subtypes—grooming, sextortion, production of CSAM, coordinated criminal networks—and where disruption is most realistic. Across 30+ platforms and 61 task forces, the same capability types surface again and again—anonymity, disappearing content, file distribution, contact discovery, trust-building at speed—regardless of which platform is in the headlines. The technology changes. The exploitation mechanics do not.

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1. Exploitation Is a Human Choice

The capabilities that make the internet valuable to hundreds of millions of legitimate users are the same capabilities offenders have learned to turn against children. Every case in this corpus begins with the same decision: a person chose to exploit a child.

Across the twenty-four years and 6,323 enforcement records this corpus spans, what technology changed was not that decision but its operational scope.

Child sexual exploitation is not a problem the internet invented. It is among the oldest documented harms to children, and the children it targets have always been vulnerable for the same reasons: age, developmental stage, and dependency on adults. What the internet changed is the capability available to the person who decides to act on that vulnerability. The offender in 1990 operated inside constraints that no longer exist in the same form. Physical proximity was required to make contact. Geography bounded victim access. Material was difficult to produce, store, and distribute. Detection risk was comparatively high at every step. In 2026, none of those constraints hold. A device in a pocket provides global reach. Anonymity is default on the most-used platforms in the world. Content can be produced, encrypted, distributed, and erased in seconds. The barriers did not fall because technology made people worse. They fell because technology raised the capability ceiling for everyone — including those who use it to harm.

This is the amplification problem, and it is distinct from causation. No platform built a feature to enable child exploitation. The features exist for hundreds of millions of legitimate users, and bad actors have learned to use the same capabilities that make the technology valuable. That is not a design failure unique to any company; it is the reality of the modern digital society. It is also why the question *which platform did this?* is less useful than the question this paper asks: *which capabilities are consistently reached for, and why do they appear across every platform in the enforcement record?*

Offenders in this corpus range from individuals who systematically identified and groomed multiple victims across years to those who acted on an available opportunity technology placed in front of them. Both types are in the enforcement record. Both made a choice. Understanding how malicious users transform platform affordances into misuse surfaces is the prerequisite for designing against them.

The landscape of child exploitation continues to evolve alongside technology. The dominant image—a predator posing as a peer, deceiving a child into a meeting

that ends in assault—is one offense type in the enforcement record. It is not the only one. Sextortion, in which offenders coerce victims through threats to expose images, emerged as a distinct and rapidly growing typology only in the last decade, made possible by the same digital infrastructure that connects everyone online. Coordinated networks of offenders operating across gaming platforms represent another. Each capability introduced has, in turn, produced offense patterns that did not previously exist at scale. Offending profiles and capabilities are not static. They expand with technology.

The harm in these cases is not abstract. Livingstone and Smith [11] document the inseparability of digital and physical risk for children: what happens on a platform shapes what happens offline, and the boundary between them is not meaningful to the child living inside it. Contact made in a chat room migrates. Material produced in one context circulates in others long after the offense ends. The platform did not manufacture predatory intent. It provided the infrastructure through which that intent found a child.

This paper asks three sequential questions about that infrastructure. Which capabilities do offenders consistently reach for, across offense types, platforms, and years? How do those affordances map to harm vectors across the technological eras this corpus spans? Where in that infrastructure does intervention become realistic? The affordance framework developed in §2 provides the analytical language: a way to ask not which platform was involved but which capability was used, and what that capability made possible. The answer turns out to be stable across three decades of platform evolution. Anonymity. Ephemerality. Distribution infrastructure. Contact discovery. Trust architecture at speed. These capabilities appear in the enforcement record because they are structurally useful: they translate platform properties into offense mechanics.

Exploitation is a human choice. The technology does not make that choice. But it shapes, in ways this paper measures at scale for the first time, the harm signatures of modern offenders.

2. The Infrastructure of Harm

2.1 Prior Work: CSEA, Platforms, and the Enforcement Ecosystem

Online child exploitation is one of the most extensively documented crime categories in the federal enforcement ecosystem. It is also one of the least studied at the case level. Understanding why requires understanding what the existing

infrastructure does, and challenges the system faces.

How ICAC Enforcement Works. The primary law enforcement response in the United States is organized through the Internet Crimes Against Children Task Force Program: 61 federally coordinated task forces representing over 5,400 federal, state, and local agencies, established in 1998 alongside the NCMEC CyberTipline and expanded continuously since. In fiscal year 2024, ICAC task forces conducted approximately 203,467 investigations and arrested more than 12,600 suspected offenders. Investigations are classified as proactive—investigators initiating contact in undercover capacity—or reactive, triggered by a tip or complaint. The FBI’s Violent Crimes Against Children program and Homeland Security Investigations operate in parallel for multi-jurisdictional cases.

The statutory framework has three main layers. The PROTECT Our Children Act of 2008 codified the ICAC program and established mandatory reporting requirements. 18 U.S.C. § 2258A requires electronic service providers to report apparent CSAM to the CyberTipline. The REPORT Act of 2024 extended those obligations to online enticement and child sex trafficking for the first time. In 2024, NCMEC received approximately 20.5 million CyberTipline reports — equivalent to 29.2 million separate incidents — from platforms operating globally. Against that volume, 12,600 arrests represents successful case resolution for a fraction of what is reported. This is not a failure of law enforcement. It is a scale problem: the volume of suspected exploitation arriving through the CyberTipline vastly exceeds the number of investigators available to act on it, and triage quality determines which offenders are caught and which victims get reached.

The Response Infrastructure. The technology ecosystem for child protection operates on three planes.

The first is content detection. PhotoDNA, developed by Microsoft in 2009, generates perceptual hashes of images and compares them against a database of known CSAM, detecting matches even after modification. It is deployed by Google, Meta, Twitter, Reddit, Discord, and dozens of other platforms. Google’s CSAI Match performs the equivalent function for video. Meta’s open-sourced PDQ and TMK+PDQF algorithms extend perceptual hashing to smaller platforms that cannot build detection infrastructure independently. Thorn’s Safer platform takes a complementary approach: perceptual hashing against a database of over 82 million verified CSAM hashes combined with an AI-based classifier designed to identify novel material that has never been hashed, with cross-platform hash-sharing that propagates newly identified CSAM across the detection ecosystem without delay. According to the Tech Coalition’s 2023 annual survey, 89% of member companies

deploy at least one image hash-matcher; 59% use video hash-matching.

This infrastructure is the backbone of CSAM detection at scale, and its effectiveness for known material is well-established [3]. A reported false positive rate below one in a trillion makes it operationally reliable. It also has four structural limits that become more significant as the threat landscape evolves. Hash-matching requires a prior identification — new material, including AI-augmented content, has no hash to match. Detection only occurs on cooperating platforms — encrypted messaging services, smaller platforms, and dark web networks are outside the detection envelope. Hash-matching is also increasingly vulnerable to adversarial attack: recent research has demonstrated that gradient-based image modifications can defeat PhotoDNA detection while leaving content visually intact. And hash-matching answers exactly one question: is this image known illegal material? It cannot answer any of the investigatively important questions — which platform capability made this offense possible, who created this material, how does this case relate to the hundreds investigated last year across the same platform.

The Behavioral Research Base. The second major body of work is empirical. The Crimes Against Children Research Center at the University of New Hampshire produced the most systematic research on online CSEA offending. Wolak et al. [18] documented the dynamics of internet-initiated sex crimes through a national law enforcement survey: who offends, how online relationships develop, how offenses progress from digital contact to physical harm. Wolak et al. [19] established that most internet-initiated sex offenses do not fit the predator-deception model that dominates prevention messaging — most offenders did not disguise their age or intentions but cultivated relationships over time, exploiting adolescent developmental vulnerability rather than deceiving victims about who they were. Wolak and Finkelhor [17] and Wolak et al. [20] characterized sextortion as an emerging and distinct offense typology with its own dynamics and victim population. Kloess et al. [9] provided a comprehensive framework for the process of online sexual grooming across internet communication platforms: prevalence, offense stages, offender characteristics, and legal responses across jurisdictions. Kloess et al. [10] mapped offense processes specifically through internet communication platforms, tracing how offenders move victims from initial contact through escalation. Livingstone and Smith [11] documented the intertwining of digital and physical risk for children at national scale. Mitchell et al. [12] characterized the law enforcement response to internet-facilitated commercial sexual exploitation using a nationally representative sample of agencies.

This body of work established the behavioral landscape of online CSEA. It did not — because the data and tools did not yet exist — analyze the relationship between

specific platform capabilities and the offense patterns those capabilities make possible across thousands of cases. The affordance lens has itself been applied to CSEA conceptually — by psychologists modeling technology-mediated abuse [15], and by criminologists theorizing the mechanics through which technological design “invites” technology-facilitated violence [21]. What no prior work has done is connect behavioral patterns to platform capabilities via an affordance-level abstraction, operationalized against enforcement case records at scale.

Operational Forensics. The third body of work is forensics. After a device is seized, investigators use enterprise forensics platforms to examine it: Nuix for large-scale e-discovery and timeline reconstruction, Magnet Axiom for mobile and cloud forensics, Cellebrite UFED for physical and logical extraction. The workflow is precise — physical images of storage media, file carving to recover deleted content and fragments, parsing of application databases and encrypted chat logs, artifact recovery from unallocated space, timeline reconstruction from access metadata and system logs. These platforms are powerful, expensive, and infrastructure-heavy. They ask what is on this phone. Not what this case shares with five hundred cases investigated last year across the same platform.

All three layers of this infrastructure carry an acute human cost. Investigators in child exploitation units routinely encounter material that has no equivalent in other areas of law enforcement. Secondary traumatic stress affects digital forensics investigators at documented rates of 40–60%, with consequences that include emotional numbing, hypervigilance, intrusive ideation, and accelerated burnout. The volume of evidence has grown as the threat landscape has expanded — a single device seizure now routinely contains hundreds of thousands of images and terabytes of cloud-synced material. The cognitive and psychological load scales accordingly. Triage tools that reduce manual exposure are not only an efficiency argument. They are a wellbeing argument.

The harm for victims in these cases does not end with the offense. Material produced circulates long after investigations close. Survivors navigate criminal proceedings, civil systems, and the ongoing reality that digital content does not disappear. Investigators carry the cases they have worked. Families carry what those cases revealed about someone they trusted. These are not incidental facts about a difficult field. They are part of why the harms this paper addresses matter. Every unworked CyberTip is a case that may go uninvestigated. Every missed cross-case pattern and unreported platform capability is an offense trajectory that could have been identified earlier. The infrastructure described here — detection, enforcement, behavioral research, operational forensics — is staffed by capable, motivated professionals working under immense constraints of scale, an evolving

threat landscape, and the sustained weight of radioactive material. The following section introduces the ontological and affordance-based framework that makes cross-case, platform-level analysis possible at corpus scale — without the direct exposure that makes engagement in this field so difficult to sustain.

2.2 What Technology Makes Possible

“Platform” is the wrong unit of analysis for this problem.

Platforms emerge, scale, and disappear. Myspace is gone. Omegle was shut down. Discord did not exist fifteen years ago. If the analytical unit is the platform, every new platform generation requires research from scratch. The offense mechanics do not restart with each platform generation. The capabilities that appear consistently in the enforcement record — anonymity, ephemerality, unmonitored communication, contact discovery, distribution infrastructure — appear on every platform, implemented differently but functioning in similar manners. The unit that holds across platform generations is affordance.

The concept of affordance originates in ecological psychology. Gibson [5] proposed that organisms do not perceive their environment as a collection of objects with fixed properties. They perceive action-possibilities: what the environment affords the organism. A door handle affords gripping. A cliff edge affords a scenic viewpoint for a hiker, a jumping point for a daredevil, and a fatal fall for the unwary. The affordance is relational: it exists between what the environment makes available and what the organism can do with it. Norman [14] translated this to design: artifacts communicate possibilities for action independent of the designer’s intent. Children can press a button before anyone explains what it does — the design affords the action. Hutchby [8] extended this to technology: artifacts have material properties that shape what users can do with them. However, affordances emerge from the interaction between what the artifact makes available and what the actor brings to it, not from the designer’s intent alone. Bucher and Helmond [2] applied the framework to social media platforms specifically, distinguishing high-level affordances — what the platform as a whole makes possible — from low-level affordances embedded in specific interface features.

The translation to this domain is direct. End-to-end encryption, designed to protect private communication for billions of legitimate users, affords unmonitored contact: communication that no platform can scan and no investigator can access without a court order. That affordance exists on WhatsApp, Signal, Telegram, and in the private message systems of every major platform. Timer-based content deletion, designed to make casual sharing feel lower-stakes, affords ephemerality: communication without a persistent record. On Snapchat, on Instagram Stories,

on WhatsApp — material sent disappears. In the enforcement record, this is evidence destruction by design. Account creation flows that require no verified identity, built to reduce onboarding friction, afford anonymity: action without accountability. File hosting and distribution infrastructure, built for sharing photos and documents, affords mass distribution of any content to anyone with a link. Location-based discovery features, built to surface nearby users for social or dating purposes, afford proximity contact with strangers. Generative AI tools, built to expand computing possibilities and revolutionize content creation, afford the production of synthetic exploitative material, the fabrication of convincing personas at scale, and the automation of contact and grooming that previously required sustained human attention — capabilities that did not exist at this level of accessibility five years ago.

Different platforms. Expanding capabilities. Same function in the enforcement record.

Platforms are not designed to be exploitation-resistant. They are designed for engagement, retention, and scale [7]. The product decisions that produce each of these affordances are not made especially for offenders, or for any specific subset of the hundreds of millions of users. The capability exists for everyone. Misuse is not a design failure in the conventional sense; rather, it is an emergent consequence of building for everyone when the population includes people who have already decided to harm.

The affordance names what the platform offers and the enforcement record provides evidence for how offenders misappropriated them. Section 3 presents both at scale.

3. Evidence at Scale: Analysis of 6,323 Exploitation Cases

3.1 Collection, Processing, and Validation

The corpus underlying this analysis consists of 6,323 publicly available Internet Crimes Against Children case records collected from 56 law enforcement sources spanning 2002 to 2026. Sources include task force press releases, agency annual reports, DOJ CEOS announcements, NCMEC public outputs, and federal court filings. All records are publicly available. No private case data, investigative files, or victim-identifying information is included in any form. The collection and analysis protocol operates under HRPO NHSR Determination #7668; this research does not contain private or personally identifiable information under

federal regulations [45 CFR 46.102(f)(1),(2)].

Source selection targeted the full range of the U.S. ICAC enforcement ecosystem: 61 task forces, federal pipelines including Army CID, Customs and Border Protection, ICE/HSI, the U.S. Secret Service, and the U.S. Marshals Service, and 2,864 unique law enforcement agencies. Collection expanded from ICAC-specific queries to broader child exploitation search terms, yielding 56 sources and a corpus representing 24 years of enforcement activity.

Processing operates through a hybrid deterministic and interpretive machine-learning pipeline across three stages.

The first stage is regex-based extraction. Deterministic rules extract structured fields from case narratives: perpetrator age, registered sex offender status, gender, and multi-defendant flags; victim age, age ranges, stated victim count, and gender after validation; relationship labels (kin, role, or stranger when not stated); charge phrases with counts and booking stage from arrest through sentencing; sentence durations; and evidence volume when quantified in text (image, video, storage, and message counts). Platform identification extracts named applications and surfaces — social media, messaging, gaming, file hosting, livestreaming, early-era chat clients, and generative AI tools when cited — alongside generic labels (online, chat, social media) when no product match is found. Technology signals are stored separately from the platform list: investigation tooling (PhotoDNA, hash-matching, CyberTipline language), anonymization infrastructure (Tor, dark web, cryptocurrency), and P2P clients.

The second stage is pattern-based classification. Each case receives topic labels (production, possession, distribution, trafficking, CSAM, AI-generated CSAM, sextortion, hands-on versus online-only, family versus stranger, international, multi-state) and severity indicators (infant, very young, under 12, sexual abuse, multiple perpetrators). Severity phrases used in triage priority scoring are extracted separately: language indicating active, ongoing, or escalating abuse is weighted in the priority model.

The third stage is ML enrichment. When the ML stack is enabled, NER via Stanford NLP (Stanza) adds organizations, locations, dates, and ages, merged with regex output under a precedence model that preserves deterministic extractions when conflicts arise. A victim-age gate drops decoy and headline ages introduced by undercover investigation language. Semantic sentence scoring attaches concept weights to each case; grooming severity tags and strong possession or AI-generation language reinforce corresponding case topics. Concept scores are retained for downstream analysis; a curated subset are merged into main fields.

Gender extraction is role-aware throughout, distinguishing victim from perpetrator gender independently to prevent conflation across multi-party cases. Agency normalization resolves over 2,800 unique agency string variants to canonical identifiers using a combination of pattern matching and NER, enabling cross-source aggregation across jurisdictions that name the same agency differently across releases.

The primary evidence base for platform harm analysis (Q1) is a candidate pool selected by automated filter, not manual case selection. A case enters the pool if its structured metadata contains at least one non-generic platform label — excluding generic placeholders (“online,” “internet,” or generic tags). This yields 1,493 candidate cases (23.6% of the corpus) spanning 54 named platforms; the remaining 76.4% lack a named platform and fall outside affordance-level analysis due to insufficient signals for supporting claims at the level of a specific platform capability. This is partly a property of the source material itself: public case documents systematically under-report platform and offense detail to avoid influencing offending behavior, compromising ongoing investigations, or revealing prosecutorial strategy (discussed in Section 7.2). Restricting Q1 to cases with a named platform trades corpus breadth for evidentiary precision as affordance-level analysis requires knowing which platform, and which of its features, was actually present.

Each candidate is processed by an evidence extractor that strips PSA and awareness-campaign boilerplate, extracts a supporting quote, assigns an offense role, and classifies evidence strength at the platform–case level. The candidate-to-evidence pipeline yields a final Q1 evidence base of 1,493 cases across 2,598 platform–case records.

Evidence strength is tiered. *Stated* records link a platform instrumentally to offense conduct in the same sentence — through an explicit construction (*through, via, using* a platform) or an offense keyword in close proximity, excluding awareness framing and enumerated app lists. *Inferred* records show platform and offense language co-occurring without instrumental linkage. *Named-only* records carry a metadata platform tag without qualifying textual support. At the case level, 639 cases (42.8%) contain at least one stated record, 98 (6.6%) carry inferred-only evidence, and 756 (50.6%) are named-only. The affordance labels applied in Q1 are a manual analytical layer over this automated extraction: the pipeline supplies quotes, roles, and tiers; affordance semantics are assigned by hand.

The full pipeline — extraction rules, classifier weights, normalization dictionaries, and reproducibility documentation — is released as open-source software under the MIT License at github.com/mrinaalr/CaseLinker.

3.2 Ontologies for Modeling Radioactive Data

CaseLinker extracts over 70,000 structured features across 10 dimensions and stores thousands of cases in a PostgreSQL database for analysis. However, tabular representations are insufficient for the questions this paper asks. A case record is not a row. It is a set of relationships: an offender reached a victim through a platform, used a capability to produce material, and distributed it through infrastructure. A relational table can store the tags, actions, and data. It cannot natively represent the structure and relationships between fields. The questions Q1’s affordance-based platform harm analysis asks — which capabilities cluster with which offense types, which platform features co-occur across cases, how offense mechanics distribute across a 24-year enforcement record — are relational questions. They require a data model that treats relationships as first-class objects.

Knowledge graphs provide that model. Each case in this corpus is represented as a directed, typed graph: nodes for entities (offenders, victims, platforms, charges, agencies), edges for relationships (`contacted_via`, `charged_under`, `investigated_by`), and typed properties for attributes. This representation enables the pattern queries Q1 requires: offense events, platforms, agencies, and actors as pieces of an interconnected system, not isolated records.

The vocabulary that models these cases is the Crimes Against Children (CAC) Ontology, shepherded by Project VIC International as an interoperable standard for representing crimes-against-children investigations. The stack is layered: gUFO provides foundational ontology primitives. UCO (Unified Cyber Ontology) models cyber investigation objects and relationships. CASE (Cyber-investigation Analysis Standard Expression) defines how investigation narratives are expressed as knowledge graphs. CAC extends that stack for crimes against children specifically — platforms, victims, offenders, investigations, and outcomes as typed entities rather than free text.

CaseLinker maps each extracted case feature to a CAC class. Platform mentions become typed `cac:Platform` nodes with semantic relationships to offense events. Statutory charges map to identifiers linked to offense type taxonomies. Perpetrator behaviors map to grooming stage classes. The mapping is explicit and auditable: every ontological assertion traces to a source extraction and a source text span. There are no black-box inferences. Structural validation is performed using SHACL (Shapes Constraint Language) against a curated subset of the CAC module stack; each case graph is validated prior to inclusion in the Q1 evidence base, and non-conforming graphs are flagged and excluded rather than silently retained.

The result is a corpus of 1,000+ SHACL-validated case graphs forming the ev-

idence base for Q1 analysis. Graph construction at corpus scale is handled by the CaseLinker MCP (Model Context Protocol) server: 34 case2cac tools for automated graph construction, validation, traversal, and export from press-release and case-record sources.

Primary-source court record representations — dockets, statements of offense, and indictments retrieved from primary federal court filings — are handled by the CASE-UCO SDK directly. The CASE-UCO SDK is a typed, multi-language library (Python, C#, Java, Rust) implementing 428+ ontology classes across the CASE/UCO standard, designed for constructing and validating JSON-LD knowledge graphs for digital forensics and cyber-investigation data. Where the case2cac tools operate over structured fields at scale, the CASE-UCO SDK is used for precision representation: extracted facts from primary federal court documents are mapped to typed ontology classes, producing ground-truth graphs that anchor corpus-level pattern analysis. The specific facts-to-graph process for each PACER case is detailed in Section 3.3.

3.3 Research Questions and Analytical Approach

Three research questions organize the empirical analysis. Each draws on a distinct evidence layer: the 6,323-case corpus for breadth, the 1,000+ validated knowledge graphs for structured pattern analysis, and four primary-source federal court records for ground-truth harm-signature anchoring.

Q1 — Platform Affordances. Which platform capabilities do offenders exploit most consistently, across case records, offense types, and the full period this corpus covers? Q1 operates at corpus scale. The 30+ platform affordance table developed in the analysis phase maps each platform in the enforcement record to its documented misuse surfaces and harm vectors, drawing on stated, inferred, and named-only evidence tiers from the extraction pipeline. The knowledge graph representation enables this analysis without reducing to frequency counts: a query across the graph can ask which capability types co-occur with which offense types, which platforms share affordance profiles, and how those profiles shift across offender typologies and agencies the corpus represents.

Q2 — Exploitation Life Cycle. How is technology implicated across the four offense subtypes — grooming and enticement, sextortion, CSAM production, and coordinated criminal networks — from first contact through prosecution? Q2 operates at the harm-signature level. The primary evidence base is four federal district court cases retrieved from the Public Access to Court Electronic Records (PACER) system and read by hand in full: *United States v. Rehman* (D.D.C. 1:23-cr-00064), a §2422(b) enticement prosecution documenting platform-mediated grooming and

relationship formation; *United States v. Amin* (D. Alaska 3:22-cr-00055), a 13-count §2252A(g) enterprise prosecution documenting sextortion at scale across more than 80 Snapchat and 40 Instagram accounts; *United States v. Herrera* (W.D. Tex. 3:25-cr-01046 and D. Alaska 3:24-cr-00091), a dual-district §2251 production case involving AI-augmented CSAM generation consolidating via Rule 20 for a single plea proceeding; and *United States v. Bermudez et al.* (E.D.N.Y. 1:25-cr-00361), a six-defendant §2252A(g) coordinated network prosecution anchored to Discord and gaming platform infrastructure. Each case was mapped to the CAC ontology independently of the automated pipeline, anchoring corpus-level patterns against primary federal court records read in full.

Q3 — Disruption. Where in the platform stack and enforcement pipeline do realistic opportunities to disrupt exploitation exist? Q3 synthesizes Q1 and Q2: given which capabilities offenders consistently reach for, and how those capabilities combine into recognizable offense mechanics, which affordances are most amenable to design-level, policy-level, or enforcement-level intervention? Q3 is the operational output of the analysis — the translation from documented pattern to actionable target.

Q1 characterizes the affordance landscape. Q2 anchors it in the court-verified mechanics of four distinct offense types. Q3 asks what can be done about it. The following three sections address each question in turn.

4. Platform Affordances and Misuse Surfaces

The enforcement record names 54 distinct platform labels across the Q1 evidence base. Analyzed individually, they appear chaotic: products from different eras, built by different companies, serving different purposes. Analyzed by affordance, they resolve into a small number of recurring signatures. A platform’s affordance profile predicts its harm profile. The same chat affordance, implemented on a platform from 2008 or one from 2025, produces the same offense pattern in the record.

Table 1 presents the platform manifest: every analyzed platform with sufficient case representation, its affordance signature, the misuse surface that offenders utilize, and the victim-facing harm vectors documented in the corpus. Three definitions discipline the table. An **affordance** is a neutral capability the platform’s design makes possible. A **misuse surface** is how an offender turns that affordance toward an offense; it is offender-facing. A **harm vector** describes what happens to a child; it is victim-facing, drawn from a fixed set established across the corpus,

and ranked by documented prevalence. Detection events (platform-reported CyberTips) are routed to disruption analysis (§6), not counted as harm. Every platform with documented harm to more than two children receives analysis: each represents real victims and a real affordance pattern.

The manifest measures *documented platform role*, not ground-truth harm frequency. Possession and distribution of CSAM dominate the record partly because they are the most detectable evidence of harm and play a significant charging role after being surfaced through the CyberTip pipeline; contact and grooming are systematically under-detected and under-documented in public records. Affordances and misuse surfaces are derived from analysis of each platform’s capabilities; harm vectors are drawn only from offenses the corpus actually documents. Platform counts are a lower bound on involvement, not a complete picture.

Table 1: Platform affordance manifest. Each platform’s defining affordance signature, the misuse surface it creates, and the victim-facing harm vectors documented in the corpus, ranked by prevalence.

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
Messaging Services		
Snapchat 126 cases	<i>Ephemerality, camera-first capture.</i> Evidence destruction by design; real-time coercion of imagery; ephemeral solicitation with reduced forensic trail.	Child contacted and exploited through ephemeral messaging Child’s abuse imagery distributed / re-shared Child groomed toward contact
Kik 124 cases	<i>Anonymity, group discovery.</i> Pseudonymous contact without traceable identity; closed-group trading; account churn to evade detection.	Child’s abuse imagery distributed / traded (re-victimization) Child contacted through anonymous messaging Child groomed toward contact
Discord 29 cases	<i>Server/community structure, persistent contact.</i> Closed servers as low-visibility spaces; coordination among offenders; sustained contact enabling coercion.	Child’s abuse imagery distributed / traded (re-victimization) Child contacted through servers and DMs Child sextorted under threat Child groomed toward contact
Telegram 8 cases	<i>[affordances pending]</i>	<i>[pending analysis]</i>
WhatsApp 6 cases	<i>[affordances pending]</i>	<i>[pending analysis]</i>

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Table 1 (continued)

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
Facebook Messenger 6 cases	[affordances pending]	[pending analysis]
IRC 6 cases	[affordances pending]	[pending analysis]
Skype 4 cases	[affordances pending]	[pending analysis]
Social Media Platforms		
Facebook 45 cases	Real-name profiles, friend graph, minimal verification. Fake/impersonation accounts; profile-mining to target; Messenger as private channel.	Child's abuse imagery distributed/re-shared Child contacted through profiles and Messenger Child groomed toward contact
Instagram 21 cases	Visual-first profiles, DM, discoverability. Visual profiles to select targets; DM coercion; disappearing messages reduce forensic trail.	Child contacted through DMs and visual profiles Child's abuse imagery distributed/re-shared Child sextorted under threat
Twitter / X 15 cases	[affordances pending]	[pending analysis]
Reddit 7 cases	[affordances pending]	[pending analysis]
TikTok 4 cases	Algorithmic discovery, DM, public-by-default. For You feed surfaces minors without active search; public videos to select targets; DM to move contact off the public feed.	Child contacted and exploited through messaging Child groomed toward contact or meeting Child solicited for explicit imagery
File Hosting Services		
Dropbox 20 cases	[affordances pending]	[pending analysis]
Google Drive 12 cases	[affordances pending]	[pending analysis]
Mega.nz 4 cases	[affordances pending]	[pending analysis]
Anonymous Chat Platforms		
Whisper 3 cases	[affordances pending]	[pending analysis]

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Table 1 (continued)

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
Omegle 2 cases	[affordances pending]	[pending analysis]
AI Services		
Gen AI 14 cases	Generative synthesis, likeness manipulation. Synthetic CSAM from prompts; nudification of real photos; model fine-tuning on a target's images.	Real child's likeness manipulated/synthesized into abuse imagery without contact Child's abuse imagery (incl. AI-generated) distributed/possessed
Additional Surfaces		
P2P (BitTorrent) 3 cases	[affordances pending]	[pending analysis]
Video Streaming 9 cases	[affordances pending]	[pending analysis]
Gaming Platforms varies	[affordances pending]	[pending analysis]
Online Dating varies	[affordances pending]	[pending analysis]

4.1 Contact and Approach Affordances

4.2 Possession and Trade Affordances

4.3 Coordination Affordances

4.4 Production Affordances: The Generative Evolution

5. Harm Signatures: How Technology Shapes Exploitation

5.1 Grooming and Enticement

5.2 Sextortion: Coercion After Contact

5.3 The Production of Abuse Material

5.4 Coordinated Criminal Enterprises

6. Disrupting Offense Mechanisms

6.1 Design-Level Interventions: Building Against Exploitation

6.2 Policy and Investigative Levers

6.3 What the Record Reveals

7. Designing Against Predation

7.1 The Foreseeable Harm Standard

7.2 Limitations and Scope

8. The Next Platform Will Be Different. The Harm Will Not.

9. Data, Code, and Public Research Artifacts

[Availability statement — to be written]

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